Lección 1

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Plan de trabajo

Organización del curso (5 min):

►Discusión del Syllabus, bibliografía, sistema de evaluación, ...

Lección (65 min):

►Repaso de Special Relativity:

UNIDAD 1. Teoría de la Relatividad Especial

1.1. Transformaciones de Galileo
1.2. Experimento de Michelson y Morley
1.3. Postulados de Einstein
1.4. Transformaciones de Lorentz, Grupo de Lorentz
1.5. Espacio-tiempo de Minkowski y vectores 4-dimensionales
1.6. Cinemática Relativista
1.7. Dinámica Relativista
1.8. Aplicaciones: la paradoja de los gemelos, efecto Doppler relativista

Discussion of syllabus (FIS-310)

UNIVERSIDAD TECNICA FEDERICO SANTA MARIA

Doctorado en Ciencias Físicas
Programa Conjunto
Universidad Técnica Federico Santa María
Pontificia Universidad Católica de Valparaíso

Asignaturas de Formación General (Obligatorias)

Nombre del curso	**MECÁNICA CLÁSICA** Código USM: FIS310 Código PUCV: FIS810
Contenidos	UNIDAD 1. Teoría de la Relatividad Especial 1.1. Transformaciones de Galileo 1.2. Experimento de Michelson y Morley 1.3. Postulados de Einstein 1.4. Transformaciones de Lorentz, Grupo de Lorentz 1.5. Espacio-tiempo de Minkowski y vectores 4-dimensionales 1.6. Cinemática Relativista 1.7. Dinámica Relativista 1.8. Aplicaciones: la paradoja de los gemelos, efecto Doppler relativista UNIDAD 2. Formulación Lagrangiana 2.1. Variables Generalizadas 2.2. Clasificación de los Sistemas Mecánicos 2.3. Elementos de Cálculo Variacional 2.4. Ecuaciones de Lagrange 2.5. Aplicaciones: movimiento en un campo central, observadores no inerciales UNIDAD 3. Teoremas de Conservación 3.1. Teorema de Noether. 3.2. Aplicaciones del teorema de Noether. UNIDAD 4. Formulación Lagrangiana de la Partícula Relativista 4.1. Lagrangiano de la Partícula Relativista 4.2. Ecuaciones de Lagrange para Partículas Relativista 4.3. Aplicaciones UNIDAD 5. Formulación Hamiltoniana 5.1. Transformaciones de Legendre 5.2. Ecuaciones de Hamilton 5.3. Ecuaciones de Hamilton de la Partícula Relativista 5.4. Corchete de Poisson 5.5. Transformaciones Canónicas 5.6. Ecuaciones de Hamilton-Jacobi 5.7. Teoría de perturbaciones

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Bibliografía	**Básica:** ▪ H. Goldstein, C. P. Poole, J. Safko, "Mecánica Clásica" (2006, San Francisco, U.S.A.: Adisson-Wesley) ▪ L.D. Landau and E. M. Lifshitz, *Mechanics.* **Recomendada:** ▪ S. T. Thornton, J. B. Marion, "Dinámica Clásica de las Partículas y Sistemas" (2003, México, D. F., México: Brooks Cole) ▪ A. L. Fetter, J. D. Walecka "Theoretical Mechanics of Particles and Continua" (2003, New York, U.S.A.: Dover Publications Inc.) ▪ Arnold, Mathematical Methods of Classical Mechanics (Springer 2nd edition 1997).

- Voy a seguir el syllabus, combinando materiales de diferentes libros.

Discusión del syllabus (FIS-426)

Syllabus de FIS-426:

1) Principios Variacionales, Lagrangianos y Hamiltonianos
   - Principio de Hamilton y Lagrangianos
   - Hamiltonianos
   - Otros principios variacionales
   - Simetrías y Leyes de Conservación (Teorema de Noether)
2) Algunas Aplicaciones
   - Fuerzas Centrales
   - Sólido Rígido
   - Pequeñas Oscilaciones
3) Sistemas Continuos y Campos
   - Mecánica de sistemas continuos como límite de sistemas de partícul
   - Formulaciones Lagrangiana y Hamiltoniana
4) Formulación Hamiltoniana General
   - Paréntesis de Poisson
   - Transformaciones canónicas
   - Variables de ángulo y acción
   - Teoría de perturbaciones
   - Caos en sistemas Hamiltonianos

.

- Los programas de FIS-345/FIS-426 se solapan en gran medida con el programa de FIS-310, pero hay algunos temas adicionales.
- Seguiremos el syllabus de FIS-310, complementándolo con los temas adicionales de FIS-345 y FIS-426.

Sistema de evaluaciones

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Modalidad de evaluación	La evaluación consiste en las tareas obligatorias que entrarán en la evaluación final, y las tareas no obligatorias que en su conjunto buscan mejorar el aprendizaje del estudiante. Se realizarán tres pruebas y se hará un examen final. La nota final estará compuesta por: Evaluación — Porcentaje Prueba 1 — 15% Prueba 2 — 15% Prueba 3 — 15% Tareas obligatorias — 15% Examen final — 40% La evaluación y los porcentajes pueden ser modificados por el profesor, informando al inicio del curso.

- Las fechas de certamenes y tareas están disponible en Aula (Tabla "**Fechas tentativas**").

¿Dudas?

Lecture plan

- Short recap of Special Relativity

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- I will follow the books of Goldstein and Landau from our bibliography, but will extend with extra materials when appropriate.
- Special Relativity (Block I) is usually discussed in books of electromagnetism because it was discovered in that context, for this reason it is absent in some books of our bibliography. For example in Landau, instead of Vol. I "*Classical Mechanics*" it appears in Vol II "*Field Theory*". For this reason **for this Block** I would recommend You some additional books on electromagnetism (check the chapters which include words "relativity" or "relativistic" in the Table of Contents): Griffiths, "*Introduction to Electrodynamics*", Marion, "*Classical electromagentic radiation*", Lorrain,Corson "*Electromagnetic fields and waves*"

Revision of Special relativity

- It is expected that You've seen some basics in undergraduate courses, so we need to recall what You remember
- Please try to recall everything You remember about Special Relativity (I need your replies to choose proper learning pace)

Control question 1

What motivated development of Special Relativity?
What was wrong with "old good" formulas from Classical mechanics?

Control question 2

Are there any experimental evidence which shows that Special Relativity is not just a trick to "solve" some problems of electromagnetism?

Control question 3

Which concepts/main results of Special Relativity differ from those of Newtonian Mechanics & Galileo's transformations ?

Control question 4

One of the postulates of the Special Relativity is that speed of light is constant ...

a) In any reference frame
b) Only in inertial reference frame

c) In any medium
d) In any medium at rest
e) Only in vacuum

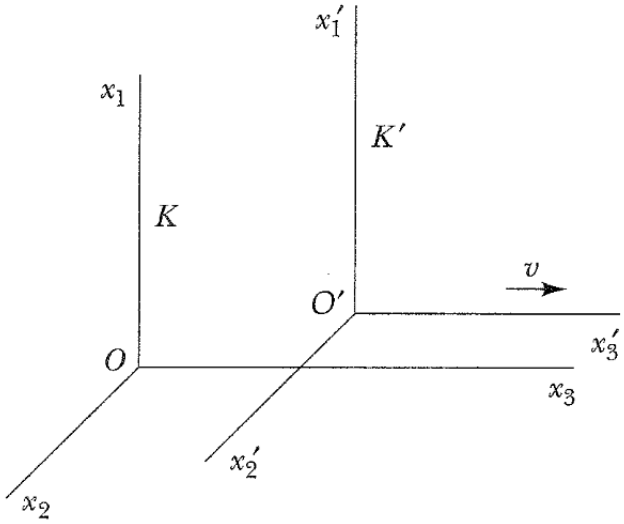
f) Only if the source of light was at rest

(Please identify all conditions which should be satisfied for validity of this postulate)

Galileo's relativity

Classical mechanics:

- All inertial reference frames are equvalent



−Time is absolute

$$t' = t$$

−Transformation of coordinates

$$\boldsymbol{x}' = \boldsymbol{x} - \boldsymbol{V}\, t$$

⇒Transformation of *distances* between points:

$$\Delta \boldsymbol{x}' = \Delta \boldsymbol{x}$$

−Transformation of velocities:

$$\boldsymbol{v} = \boldsymbol{v}' + \boldsymbol{V}$$

−Transformation of forces (recall $\boldsymbol{F} = m\,\boldsymbol{a}$):

$$\boldsymbol{F} = \boldsymbol{F}'$$

Electromagnetism vs Galileo's relativity

Maxwell's equations are incompatible with Galileo's principle:

- EM wave equation is covariant (=doesn't change its form) in any inertial frame

$$\frac{1}{c^2}\frac{\partial^2 \boldsymbol{E}}{\partial t^2} - \Delta \boldsymbol{E} = 0,$$
$$\frac{1}{c^2}\frac{\partial^2 \boldsymbol{B}}{\partial t^2} - \Delta \boldsymbol{B} = 0,$$

Same velocity c in any frame, in contradiction with Galileo's

$$\boldsymbol{v}_1 = \boldsymbol{v}_2 + \boldsymbol{V}_{\text{frame}}$$

Note that similar equations appear for acoustic waves in mechanics of fluids and elastic medium, but they were found in the medium rest frame, so there the problem does not exist !!!

Electromagnetism vs Galileo's relativity

- The force between two charges which move with the same constant velocity in lab frame

$$\boldsymbol{F} = q\left(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\right),$$

- Results for electric and magnetic fields in moving frame:

$$\boldsymbol{E} = \frac{q}{4\pi\varepsilon_0} \frac{1 - v^2/c^2}{\left(1 - v^2 \sin^2\theta/c^2\right)^{3/2}} \frac{\hat{\boldsymbol{R}}}{R^2}$$

$$\boldsymbol{B} = \frac{q}{4\pi\varepsilon_0} \frac{1 - v^2/c^2}{\left(1 - v^2 \sin^2\theta/c^2\right)^{3/2}} \frac{\boldsymbol{v} \times \hat{\boldsymbol{R}}}{R^2}$$

where θ is angle between $\hat{\boldsymbol{R}}$ and $\boldsymbol{v}$, and $\boldsymbol{R} = \boldsymbol{r} - \boldsymbol{v}t$ (*instant, not* retarded separation).

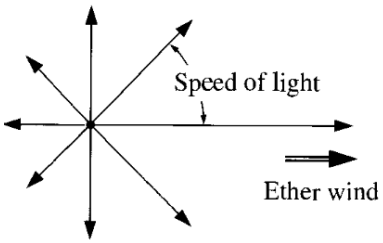
- In the rest frame we get

$$\boldsymbol{F} = \boldsymbol{F}_{\text{Coulomb}} = \frac{q_1 q_2}{4\pi\varepsilon_0} \frac{\hat{\boldsymbol{R}}}{R^2}$$

$\Rightarrow$different force between electric charges in different frames, contradicts Galileo's result $\boldsymbol{F} = \boldsymbol{F}'$

Attempts to reconcile EM with Galileo's relativity

- Early attempts (**all failed**!):
- ▶ **Modify equations of Maxwell** (add more terms to restore covariance)
- – Numerous checks, no deviations were detected.
- ▶ **ether** (fictitious invisible substance which we do not feel)
- – EM field⇒strain of ether (Maxwell's strain tensor⇒"strain of ether")
- – ether breaks explicitly the equivalence of different frames
- –motion of ether⇒"ether wind", should lead to observable consequences
  - * Example: speed of light should depend on motion of the ether
- – Attempts to measure light speed difference in different direction-zero results



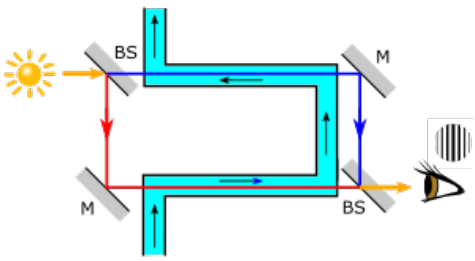
Attempts to measure the speed of light and ether drag

- Major challenge: speed $c \approx 3 \times 10^8 \mathrm{m}/s \gg$ speeds of macroscopic objects

Main idea:

– Use interference of light rays which travel via different paths, in different mediums. If the motion of the medium could affect the velocity of light, then interference patters should shift as a function of velocity

Typical experiments of that period:

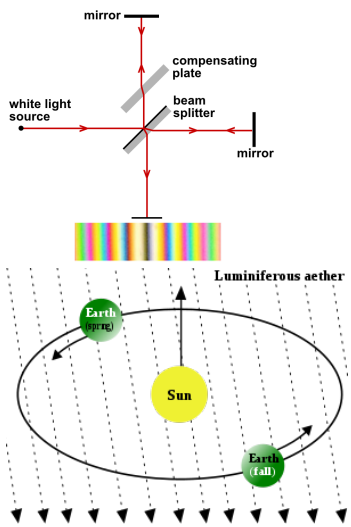


- Fizeau experiment (1851, repeated in different modifications).
  – Tried to detect if the moving water can drag ether and in this way affect the velocity of light.
  – Galileo's theory could not explain the results, $v_{\mathrm{light}} \neq (c/n) \pm v_{\mathrm{water}}$. At the same time, $v_{\mathrm{light}} \neq c/n$.

– In SR the effect is interpreted via modification of electric and magnetic polarizabilities of material in moving frame ($\boldsymbol{D} \neq \varepsilon\varepsilon_0 \boldsymbol{E}$, $\boldsymbol{B} \neq \mu\mu_0 \boldsymbol{H}$), so $v_{\mathrm{light}} \neq c/n$ (can use $v_{\mathrm{light}} = c/n$ only in rest frame of water). In any other frame should use Einstein's velocity addition formula.

– Shift of interference picture $\sim \mathcal{O}(v/c)$ [linear]

Michelson-Morley interferometer (1887)



- Tried to detect if the relative motion of the observer (interferometer) w.r.t. ether affects the speed of light
  - Assume that L_1 and L_2 are the lengths of the arms of the interferometer, which moves in horizontal direction with velocity v_x with respect to ether. Then the corresponding phase shift *according to Galileo's theory* is given by

$$\Delta\varphi = \omega\left(\frac{L_1}{c-v_x} + \frac{L_1}{c+v_x} - \frac{2\,L_1}{c}\right) =$$
$$= \frac{\omega L_1}{c}\left(\frac{1}{1-v_x/c} + \frac{1}{1+v_x/c} - 2\right)$$
$$= \frac{2L_1}{\lambda}\left(\frac{v_x^2/c^2}{1-v_x^2/c^2}\right) \sim \mathcal{O}\left(v_x^2/c^2\right)$$

[quadratic, second-order]

  - Expected: Should have periodic oscillations of the phase shift due to rotation of the Earth (change of orientation of $\boldsymbol{v}$)
  - Experiment: null result, no shift (within errors of experiment). There is no ether drag, speed of light $c = \mathrm{const}$

Electromagnetism vs Galileo's relativity

Theoretical developments:

−We know that Cartesian components of $\boldsymbol{E}$, $\boldsymbol{B}$ in vacuum should satisfy

$$\frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} - \Delta\psi = 0, \qquad (1)$$

−Assume that a plane EM wave moves just in 1 dimension in direction $\hat{\boldsymbol{x}}$.

$$\frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} - \frac{\partial^2\psi}{\partial x^2} = 0, \qquad (1')$$

Note that as of now we don't know how $\boldsymbol{E}$, $\boldsymbol{B}$ transform from one frame to another

−However, if we assume that $\boldsymbol{E}$, $\boldsymbol{B}$ transform linearly through each other,

$$E_\alpha \to K_{\alpha\beta}E_\beta + L_{\alpha\beta}B_\beta, \quad B_\alpha \to M_{\alpha\beta}E_\beta + N_{\alpha\beta}N_\beta,$$

where K, L, M, N are some coordinate-independent matrices, then, taking into account that all components of $\boldsymbol{E}$, $\boldsymbol{B}$ should satisfy the same equation (1'), can conclude that in new frame $\boldsymbol{E}$, $\boldsymbol{B}$ also must satisfy (1')

Demonstrate explicitly that (1') is not covariant under Galileo's transformations, *i.e.* if it has this form in one reference frame, it will get additional terms in another frame

Electromagnetism vs Galileo's relativity

$$\frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} - \frac{\partial^2\psi}{\partial x^2} = 0, \qquad (1')$$

–Time is absolute, transformation of time coordinates from one frame to another:

$$t' = t, \quad \boldsymbol{x}' = \boldsymbol{x} - \boldsymbol{V}\, t$$

$$\frac{\partial}{\partial x} = \frac{\partial x'}{\partial x}\frac{\partial}{\partial x'} = \frac{\partial}{\partial x'}$$

$$\frac{\partial}{\partial t} = \frac{\partial t'}{\partial t}\frac{\partial}{\partial t'} + \frac{\partial x'}{\partial t}\frac{\partial}{\partial x'} = \frac{\partial}{\partial t'} - v\frac{\partial}{\partial x'}$$

$$\frac{\partial^2}{\partial t^2} = \left(\frac{\partial}{\partial t'} - v\frac{\partial}{\partial x'}\right)^2 = \frac{\partial^2}{\partial t'^2} + v^2\frac{\partial}{\partial x'^2} - 2v\frac{\partial}{\partial t'}\frac{\partial}{\partial x'}$$

$$\frac{1}{c^2}\frac{\partial^2\psi}{\partial t'^2} - \frac{\partial^2\psi}{\partial x'^2} + \frac{v^2}{c^2}\frac{\partial\psi}{\partial x'^2} - \frac{2v}{c^2}\frac{\partial}{\partial t'}\frac{\partial}{\partial x'}\psi = 0$$

–due to additional terms (in red) the covariance is broken, wave equation is NOT covariant *w.r.t. Galileo's transformation.*

Electromagnetism vs Galileo's relativity

Theoretical developments
(Lorentz, 1899-1904):

Assume that a plane EM wave moves just in 1 dimension in direction $\hat{\boldsymbol{x}}$. Let's try to find (mathematically) a linear transformation $(x,\, t) \leftrightarrow (x',\, t')$ which keeps the wave equation

$$\frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} - \frac{\partial^2\psi}{\partial x^2} = 0, \quad (1)$$

covariant (*i.e.* has the same form in all reference frames)

Observation: The light moves with *the same* velocity in all reference frames, so its front should satisfy

$$\boldsymbol{x}^2 - c^2 t^2 = (\boldsymbol{x}')^2 - c^2 (t')^2 = 0 \quad (2)$$

Let's introduce a variable $x_0 = ic\,t$. In this case the equation (2) turns into

$$x_0^2 + \boldsymbol{x}^2 = (x_0')^2 + (\boldsymbol{x}')^2 = 0 \quad (3)$$

-equation for the norm (absolute value) of vector $\vec{X} = (x_0,\, \boldsymbol{x})$ in 4D space. Recall that only orthogonal transformations keep the value of vector invariant.

Lorentz transformations

$$x_0^2 + \boldsymbol{x}^2 = (x_0')^2 + (\boldsymbol{x}')^2 = 0 \quad (3)$$

-equation for the absolute value of vector $\vec{X} = (x_0, \boldsymbol{x})$ in 4D space. Recall that only orthogonal transformations keep the value of vector invariant.

$$x' = \cos\alpha\, x + \sin\alpha\, ict, \quad ict' = -\sin\alpha\, x + \cos\alpha\, ict \quad (4)$$

From (4): x' is real, so $\sin\alpha$ must have only imaginary part. Recall that

$$\cos(i\eta) = \cosh\eta, \quad \sin(i\eta) = i\,\sinh\eta$$

so we may rewrite (4, 5) in a form

$$x' = \cosh\eta\, x - \sinh\eta\, ct, \quad ct' = -\sinh\eta\, x + \cosh\eta\, ct \quad (5)$$

Compare with Galileo's formulas ($v \ll c$):

$$\sinh\eta \approx \frac{v}{c}, \quad \cosh\eta \approx 1$$

But $\cosh^2\eta - \sinh^2\eta \equiv 1$, so need to generalize

$$\sinh\eta = \frac{v/c}{\sqrt{1 - v^2/c^2}}, \quad \cosh\eta = \frac{1}{\sqrt{1 - v^2/c^2}}$$

Lorentz transformations (3+1)

Generalization to 3+1 dimensional case:

$$x_0^2 + \boldsymbol{x}^2 = (x_0')^2 + (\boldsymbol{x}')^2 = \mathrm{const}$$

We expect that transformation of components of vector $\vec{X} = (x_0, \boldsymbol{x})$ should be linear, akin to rotations in ordinary 3D space:

$$X^\mu \to \Lambda^\mu_{\ \nu} X^\nu$$

where $\Lambda^\mu_{\ \nu}$ are some matrices. If we had ordinary 4D Euclidean space, we would conclude that matrices Λ are orthogonal,

$$\Lambda^{\mathrm{T}} \cdot \Lambda = 1$$

However, now we should be careful because $x_0 = ict$, so some "rotation angles" are purely imaginary. We'll need some additional mathematical concept to be able to formulate carefully conditions on $\Lambda^\mu_{\ \nu}$.

Minkowsky space

Some mathematical terminology:

Minkowski space

n-dimensional space in which the norm of 4D vector is defined as

$$|a|^2 = a_0^2 - a_1^2 - a_2^2 - a_3^2$$

(some authors use notation $a_4 \equiv a_0$)

- In some books might be defined as

$$|a|^2 = -a_0^2 + a_1^2 + a_2^2 + a_3^2$$

(set of signs is called signature, $+ - - -$ or $- + + +$)

Mathematical generalization:

Pseudo-Euclidean space

n-dimensional vector space in which the norm of vector is defined as

$$|a|^2 = a_1^2 + ... + a_k^2 - a_{k+1}^2 - ... - a_n^2$$

Some standard notations & definitions:

- We'll use notation a_μ (greek index) for 4-vector $a^\mu = (a_0, \boldsymbol{a})$, and a_i (latin index) for spatial components
- **Metrics tensor: introduce matrix** $\eta_{\mu\nu} = \mathrm{diag}\,(1, -1, -1, -1)$ **and rewrite** $|a|$ **in the form**

$$|a|^2 \equiv \sum_{\mu\,\nu} \eta_{\mu\nu} a^\mu a^\nu$$

- We'll call a vector with components $a_\mu \equiv \eta_{\mu\nu} a^\mu$a covariant, and vector with components a^μcontravariant. In case of Euclidean space ($\eta_{\mu\nu} = \delta_{\mu\nu}$) they just coincide, but in Minkowski space they differ by sign in front of a_i and thus transform differently.
- Einstein's implied summation over dummy (repetitive index):

$$|a|^2 = a_\mu a^\mu \equiv \sum_{\mu=0}^{3} a_\mu a^\mu = \sum_{\mu,\nu} \eta_{\mu\nu} a^\mu a^\nu$$

Minkowsky space

Some standard notations & definitions:

- Transformation of vectors and covectors is done with different matrices:

$$x_\mu \to \Lambda_\mu^{\ \nu} x_\nu$$

$$x^\mu \to \Lambda^\mu_{\ \nu} x^\nu$$

with Minkowskian orthogonality

$$\Lambda^\alpha_{\ \mu} \Lambda_\alpha^{\ \nu} = \delta^\nu_\mu, \qquad \eta^{\alpha\beta} \Lambda_{\alpha,\,\mu} \Lambda_{\beta,\nu} = \eta_{\mu\nu},$$

Note that

$$\Lambda_\mu^{\ \alpha}(\boldsymbol{v}) = \left[\Lambda^{-1}(\boldsymbol{v})\right]^\alpha_{\ \mu} \equiv \Lambda^\alpha_{\ \mu}(-\boldsymbol{v})$$

Control question

Assume that we work in $n+1$ dimensional pseudoeuclidean space with metrics

$$|a|^2 = a_1^2 - a_2^2 - \ldots - a_{n+1}^2$$

– How many parameters are needed in order to parametrize the matrix Λ^{μ}_{ν}?

– How many parameters are needed in order to parametrize the pure spacial rotations in n-dimensional subspace?

– The difference between the two previous responses corresponds to a number of pure boosts. Does it coincide with a number of components of velocity $\boldsymbol{v}$ in this pseudoeuclidean space?

Lorentz transformations

$$\frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} - \frac{\partial^2\psi}{\partial x^2} = 0, \quad (1)$$

$$x' = \cosh\eta\, x - \sinh\eta\, ct, \quad ct' = -\sinh\eta\, x + \cosh\eta\, ct \quad (2)$$

$$\sinh\eta = \frac{v/c}{\sqrt{1 - v^2/c^2}}, \quad \cosh\eta = \frac{1}{\sqrt{1 - v^2/c^2}}$$

So far we found (2) "indirectly", from condition that EM wave propagates with the same speed of light $c = \mathrm{const}$. Let's demonstrate explicitly that (1) is covariant under Lorentz transformations

$$\frac{\partial}{\partial x} = \frac{\partial x'}{\partial x}\frac{\partial}{\partial x'} + \frac{\partial t'}{\partial x}\frac{\partial}{\partial t'} = \cosh\eta\,\frac{\partial}{\partial x'} - \frac{\sinh\eta}{c}\frac{\partial}{\partial t'}$$

$$\frac{\partial}{\partial t} = \frac{\partial x'}{\partial t}\frac{\partial}{\partial x'} + \frac{\partial t'}{\partial t}\frac{\partial}{\partial t'} = -c\,\sinh\eta\,\frac{\partial}{\partial x'} + \cosh\eta\,\frac{\partial}{\partial t'}$$

$$\Rightarrow \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} = \frac{1}{c^2}\left(\frac{\partial}{\partial t'}\right)^2 - \left(\frac{\partial}{\partial x'}\right)^2$$

i.e. (1) remains covariant

Electromagentism before Special relativity

Short summary of EM results (theory and experiment, 1905)

- ▶Maxwell's equations (ME) describe all EM very well
  - no signs of extra terms, motion of ether
  - Predict that speed of light $c = \mathrm{const}$
  - ... confirmed by all experiments
- ▶ME are incompatible with Galileo's transformation of coordinates
- ▶Invariant w.r.t. some (strange, purely mathematical) Lorentz transformations:

$$x' = \cosh\eta\, x - \sinh\eta\, ct, \quad ct' = -\sinh\eta\, x + \cosh\eta\, ct \quad (1)$$

$$\sinh\eta = \frac{v/c}{\sqrt{1 - v^2/c^2}}$$

Generalization to 3+1 dimensional case: A linear transformation

$$X^\mu \to \Lambda^\mu_{\ \nu} X^\nu$$

which does not change the Minkowskian norm (absolute value) of the 4-vector $\vec{X} = (x_0, \boldsymbol{x}) = (ct, \boldsymbol{x})$

$$X^2 = x_0^2 - \boldsymbol{x}^2 = \mathrm{const}$$

–If we redefine the variable $x_4 = ict$, will get the usual Euclidean norm (up to a sign)

$$X^2 = -\left(x_4^2 + \boldsymbol{x}^2\right)$$

Minkowskian orthogonality

$$\Rightarrow \Lambda^\alpha_{\ \mu}\Lambda_\alpha^{\ \nu} = \delta^\nu_\mu, \qquad \eta^{\alpha\beta}\Lambda_{\alpha,\,\mu}\Lambda_{\beta,\nu} = \eta_{\mu\nu},$$

Note that

$$\Lambda_\mu^{\ \alpha}(\boldsymbol{v}) = \left[\Lambda^{-1}(\boldsymbol{v})\right]^\alpha_{\ \mu} \equiv \Lambda^\alpha_{\ \mu}(-\boldsymbol{v})$$

Control question

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Electromagentism vs Special relativity

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– Predict that speed of light $c = \mathrm{const}$

–... confirmed by all experiments

►ME are incompatible with Galileo's transformation of coordinates

►Invariant w.r.t. Lorentz transformations:

$$x' = \cosh\eta\, x - \sinh\eta\, ct$$
$$ct' = -\sinh\eta\, x + \cosh\eta\, ct$$
$$y' = y, \quad z' = z$$

$$\sinh\eta = \frac{v/c}{\sqrt{1 - v^2/c^2}}$$

(though this was considered as mathematical transform)

Albert Einstein, 1905:

The principle of relativity

The laws of physics apply (are equivalent) in all inertial reference frames

►Generalization of Galileo's principle from mechanics to all possible interactions

►Eliminates the concept of ether

The universal speed of light

The speed of light in vacuum is the same in all reference frames.

►All special relativity is based on these principles

►The second postulate effectively replaces Galileo's transform with Lorentz transformations

Minor technical comments

The principle of relativity

The laws of physics apply (are equivalent) in all **inertial** reference frames

- This principle is valid only in inertial frames. It is NOT valid if we consider non-inertial frame (accelerated, rotating, etc).

The universal speed of light

The speed of light in vacuum is the same in all reference frames.

- Valid only for the speed of light **in vacuum**. Inside matter (dielectric), the speed of light

$$v = c/n < c,$$

and is NOT constant (depends on frame, velocity of dielectric, ...). We'll see a bit later in detail what happens there (and will see in detail Fizeau's experiment).

Special relativity: physical consequences

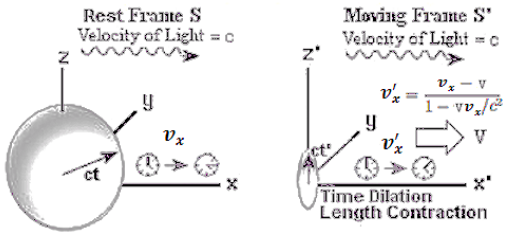
- Lorentz transformations:

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}} \quad (4')$$

$$ct' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}} \quad (5')$$

$$y' = y, \quad z' = z$$

$\eta \equiv \operatorname{arctanh}(v/c)$-***rapidity***.



– Distance between moving objects depends on frame, contracts in longitudinal direction:

$$\Delta x' = \frac{\Delta x}{\sqrt{1 - v^2/c^2}} \neq \Delta x$$

$$\Delta y' = \Delta y, \quad \Delta z' = \Delta z$$

– Time is not absolute, depends on frame *(recall "synchronization" exercise from General Physics courses)*

▷... time intervals are not constant

▷... simultaneity of events in different points is not absolute

– However, such events cannot be related by causality (would imply propagation of signal with velocity $v > c$)

– Lorentz boost cannot change time ordering for events which are related by causality.

Interval (Lorentz invariant)

Interval between two events (x_1, t_1) and (x_2, t_2) is defined as

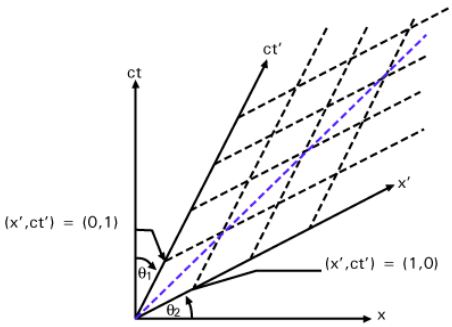
$$I \equiv c^2 \Delta t^2 - \Delta \boldsymbol{x}^2,$$

$$\Delta \boldsymbol{x} \equiv \boldsymbol{x}_2 - \boldsymbol{x}_1,$$

$$\Delta t = t_2 - t_1.$$

Intervals vs Lorentz transformations

Assume we have 2 events, A and B separated by some interval. We'll choose reference frame in which A happens at moment $t = 0$ at the origin of coordinates ($x = 0$), and B at some later time

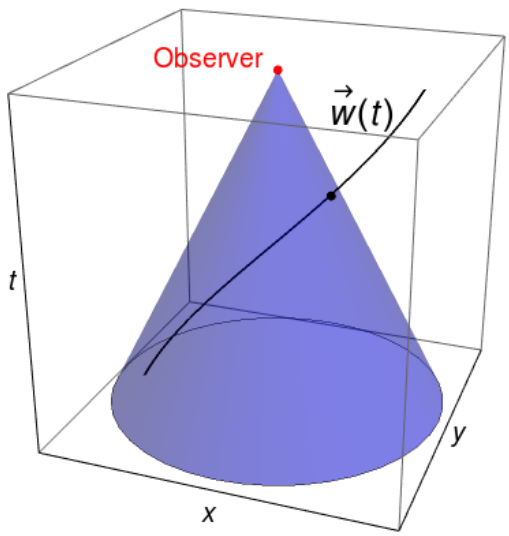


- For any point above blue dashed line ($I > 0$) can make boost after which axis ct will pass through it, *i.e.* the event will happen in the origin of cooridnates
- For any point below blue dashed line ($I < 0$) can make boost after which axis $\hat{x}$ will pass through it, *i.e.* the event B will happen at moment $t = 0$, simultaneuosly with A. We can even choose frame in which B happened before A.
- Any point on the blue dashed line ($I = 0$) remains there after any boost, *i.e.* the light-like separation is invariant in all frames.

Interval vs Causality

For observer making experiment at $(t, \boldsymbol{x})$:

1) All points inside the light cone going to the past, are separated by time-like intervals, might be related by causality/affect his measurents
2) All points outside the cone (but with $t < 0$) are separated by space-like intervals, cannot be related by causality or affect his measurents. May choose reference frame in which they happen *after* the measurement.



Control problem

►Lorentz transformations (aka "boost"):

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}},$$
$$ct' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}},$$
$$y' = y, \quad z' = z$$

Inverse Lorentz transformations may be found changing direction of velocity

$$\boldsymbol{v} \to -\boldsymbol{v}$$

:

$$x = \frac{x' + v\,t'}{\sqrt{1 - v^2/c^2}},$$
$$ct = \frac{t' + \frac{v\,x'}{c^2}}{\sqrt{1 - v^2/c^2}},$$
$$y' = y, \quad z' = z$$

Suggest how we can generalize the Lorentz transformation for arbitrary velocity $\boldsymbol{v}$, whose direction is **not** collinear to $\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{y}}$, $\hat{\boldsymbol{z}}$

Control problem (reply)

Write the Lorentz transformation for arbitrary $\boldsymbol{v}$ not collinear to $\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{y}}$, $\hat{\boldsymbol{z}}$

We need to split the vector $\boldsymbol{x}$ into parts collinear and orthogonal to $\boldsymbol{v}$,

$$\boldsymbol{x}_{||} = (x \cdot \hat{v})\, \hat{\boldsymbol{v}} = \frac{(x \cdot v)\, \boldsymbol{v}}{v^2},$$
$$\boldsymbol{x}_{\perp} = \boldsymbol{x} - \boldsymbol{x}_{||} = \boldsymbol{x} - \frac{(x \cdot v)\, \boldsymbol{v}}{v^2}$$

▷ Apply Lorentz transformation for collinear and $\perp$ parts:

$$\boldsymbol{x}'_{||} = \gamma\left(\boldsymbol{x}_{||} - \boldsymbol{v}t\right) = \gamma\left(\frac{(x \cdot v)\, \boldsymbol{v}}{v^2} - \boldsymbol{v}t\right)$$
$$\boldsymbol{x}'_{\perp} = \boldsymbol{x}_{\perp} = \boldsymbol{x} - \frac{(x \cdot v)\, \boldsymbol{v}}{v^2}$$

$$\boldsymbol{x}' = \boldsymbol{x}'_{||} + \boldsymbol{x}'_{\perp} = \boldsymbol{x} + \boldsymbol{v}\left[\frac{\boldsymbol{x} \cdot \boldsymbol{v}}{v^2}(\gamma - 1) - \gamma t\right]$$
$$t' = \gamma\left(t - \frac{\boldsymbol{x} \cdot \boldsymbol{v}}{c^2}\right)$$

Lorentz boosts

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \qquad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}}, \quad y' = y, \quad z' = z$$

Generalization for boosts in arbitrary direction:

$$\boldsymbol{x}' = \boldsymbol{x}'_{||} + \boldsymbol{x}'_{\perp} = \boldsymbol{x} + \boldsymbol{v}\left[\frac{\boldsymbol{x}\cdot\boldsymbol{v}}{v^2}(\gamma - 1) - \gamma t\right], \quad t' = \gamma\left(t - \frac{\boldsymbol{x}\cdot\boldsymbol{v}}{c^2}\right)$$

Can rewrite in compact matrix form: $x'^{\mu} = \Lambda^{\mu}{}_{\nu}(\boldsymbol{v})\, x^{\nu}$

$$\Lambda^{\alpha}{}_{\mu}\Lambda_{\alpha}{}^{\nu} = \delta^{\nu}_{\mu}, \qquad \eta_{\alpha\beta}\Lambda^{\alpha}{}_{\mu}\Lambda_{\nu}{}^{\beta} = \eta_{\mu\nu}, \qquad \Lambda_{\mu}{}^{\alpha}(\boldsymbol{v}) = \left[\Lambda^{-1}(\boldsymbol{v})\right]^{\alpha}{}_{\mu} \equiv \Lambda^{\alpha}{}_{\mu}(-\boldsymbol{v})$$

where $\Lambda^{\mu}{}_{\nu}(\boldsymbol{v})$ is the matrix which corresponds to Lorentz boost (akin to rotation matrix for rotations):

$$\Lambda^{\mu}_{\nu} = \begin{pmatrix} \gamma & -v_x\,\gamma & -v_y\,\gamma & -v_z\,\gamma \\ -\frac{v_x\,\gamma}{c^2} & 1+\frac{\bar{\gamma}v_x^2}{v^2} & \frac{\bar{\gamma}v_x v_y}{v^2} & \frac{\bar{\gamma}v_x v_z}{v^2} \\ -\frac{v_y\,\gamma}{c^2} & \frac{\bar{\gamma}v_x v_y}{v^2} & 1+\frac{\bar{\gamma}v_y^2}{v^2} & \frac{\bar{\gamma}v_y v_z}{v^2} \\ -\frac{v_z\,\gamma}{c^2} & \frac{\bar{\gamma}v_x v_z}{v^2} & \frac{\bar{\gamma}v_y v_z}{v^2} & 1+\frac{\bar{\gamma}v_z^2}{v^2} \end{pmatrix},$$

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}, \qquad \bar{\gamma} = \gamma - 1$$

$$\text{for boost in direction } \hat{\boldsymbol{x}}: \qquad \Lambda^{\mu}_{\nu}(v\,\hat{\boldsymbol{x}}) = \begin{pmatrix} \gamma & -v\,\gamma & 0 & 0 \\ -\frac{v\,\gamma}{c^2} & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

- Inverse Lorentz transformations: just change $\boldsymbol{v} \to -\boldsymbol{v}$

Lorentz boosts

$$\Lambda^{\mu}{}_{\nu}(\boldsymbol{v}) = \begin{pmatrix} \gamma & -v_x\,\gamma & -v_y\,\gamma & -v_z\,\gamma \\ -\frac{v_x\,\gamma}{c^2} & 1+\frac{\bar{\gamma}\,v_x^2}{v^2} & \frac{\bar{\gamma}\,v_x\,v_y}{v^2} & \frac{\bar{\gamma}\,v_x\,v_z}{v^2} \\ -\frac{v_y\,\gamma}{c^2} & \frac{\bar{\gamma}\,v_x\,v_y}{v^2} & 1+\frac{\bar{\gamma}\,v_y^2}{v^2} & \frac{\bar{\gamma}\,v_y\,v_z}{v^2} \\ -\frac{v_z\,\gamma}{c^2} & \frac{\bar{\gamma}\,v_x\,v_z}{v^2} & \frac{\bar{\gamma}\,v_y\,v_z}{v^2} & 1+\frac{\bar{\gamma}\,v_z^2}{v^2} \end{pmatrix},$$

$$\gamma = \frac{1}{\sqrt{1-v^2/c^2}}, \qquad \bar{\gamma} = \gamma - 1$$

- If we set $c = 1$ (effectively use variable $x^0 \equiv ct$), can see that for <u>pure boosts</u> the matrix $\Lambda^{\mu}{}_{\nu}(\boldsymbol{v})$ is <u>always</u> symmetric (!).
- Inverse Lorentz transformations: just change $\boldsymbol{v} \to -\boldsymbol{v}$

$$\text{for boost in direction } \hat{\boldsymbol{x}}: \qquad \Lambda^{\mu}_{\nu}(v\,\hat{\boldsymbol{x}}) = \begin{pmatrix} \gamma & -v\,\gamma & 0 & 0 \\ -\frac{v\,\gamma}{c^2} & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\Rightarrow x' = \frac{x - vt}{\sqrt{1-v^2/c^2}}, \qquad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1-v^2/c^2}}, \quad y' = y, \quad z' = z$$

Lemma

Assume that we make two consecutive Lorentz boosts with velocities $v_{1,2}$ in direction $\hat{x}$. The combined transformation is also a boost in direction $\hat{x}$ with rapidity

$$\eta = \eta_1 + \eta_2 = \operatorname{arctanh}\left(\frac{v_1}{c}\right) + \operatorname{arctanh}\left(\frac{v_2}{c}\right)$$

(i.e. rapidities "sum up", as angles in case of rotation around the same axis)

Lemma (proof)

We found earlier that Lorentz transofrmation in direction $\hat{x}$ is given by

$$\begin{aligned} x' &= \cosh\eta\, x - \sinh\eta\, ct \\ ct' &= -\sinh\eta\, x + \cosh\eta\, ct \\ y' &= y, \quad z' = z \end{aligned}$$

Let's disregard coordinates y, z which transform trivially and rewrite this transformation as

$$\begin{pmatrix} x' \\ ct' \end{pmatrix} = \begin{pmatrix} \cosh\eta & -\sinh\eta \\ -\sinh\eta & \cosh\eta \end{pmatrix} \begin{pmatrix} x \\ ct \end{pmatrix}$$

For superposition of 2 boosts have

$$\begin{aligned} \begin{pmatrix} x' \\ ct' \end{pmatrix} &= \begin{pmatrix} \cosh\eta_2 & -\sinh\eta_2 \\ -\sinh\eta_2 & \cosh\eta_2 \end{pmatrix} \begin{pmatrix} \cosh\eta_1 & -\sinh\eta_1 \\ -\sinh\eta_1 & \cosh\eta_1 \end{pmatrix} \begin{pmatrix} x \\ ct \end{pmatrix} \\ &= \begin{pmatrix} \cosh\eta & -\sinh\eta \\ -\sinh\eta & \cosh\eta \end{pmatrix} \begin{pmatrix} x \\ ct \end{pmatrix} \end{aligned}$$

where $\eta = \eta_1 + \eta_2$ and we simply multiplied 2 colored matrices, using

$$\begin{aligned} \cosh(\eta_1 + \eta_2) &= \cosh\eta_1 \cosh\eta_2 + \sinh\eta_1 \sinh\eta_2, \\ \sinh(\eta_1 + \eta_2) &= \sinh\eta_1 \cosh\eta_2 + \cosh\eta_1 \sinh\eta_2, \end{aligned}$$

Control problem 2

Problem 2

Assume that we make two consecutive Lorentz boost in directions $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$. Demonstrate, multiplying Lorentz boost matrices

$$(\Lambda_1)^{\mu}_{\nu} \equiv \Lambda^{\mu}_{\nu}\,(v\,\hat{\boldsymbol{x}}), \quad (\Lambda_2)^{\mu}_{\nu} \equiv \Lambda^{\mu}_{\nu}\,(u\,\hat{\boldsymbol{y}})\,,$$

that the order of multiplication is important, *i.e.* boosts are not commutative ($\Lambda_1 \cdot \Lambda_2 \neq \Lambda_2 \cdot \Lambda_1$). Also, prove that in both cases the result of multiplication ($\Lambda_1 \cdot \Lambda_2$ or $\Lambda_2 \cdot \Lambda_1$) is not pure boost.